

POKHARA UNIVERSITY

Level: Bachelor

Semester: Fall

Year : 2023

Programme: BE

Full Marks:100

Course: Digital Logic (New)

Pass Marks: 45

Time :3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What are advantages and disadvantages of analog and digital systems? Explain. 5
b) Determine the value of base x if $(50)_x = (203)_4$. 5
c) What is weighted and non-weighted code? Why Gray code is called reflected code? 5
2. a) Simplify the given function using K-Map. Also realize the simplified logic circuit using NAND gate only. 8
 $F(A, B, C, D) = \sum_m(0, 2, 6, 8, 10) + d = \sum_m(11, 12, 13, 14, 15)$
b) Construct a full subtractor circuit using two half subtractors. Also verify your answer. 7
3. a) Design a combinational circuit that converts 4-bit Excess-3 code into Gray code. 8

OR

Design a combinational circuit that converts a decimal digit from the 2421 code to BCD

- b) State and prove De Morgans theorem. Construct $F = AC + B'D$ only using universal gates. 7
4. a) Design and implement BCD to decimal converter circuit using 3-8 linedecoder. 8

OR

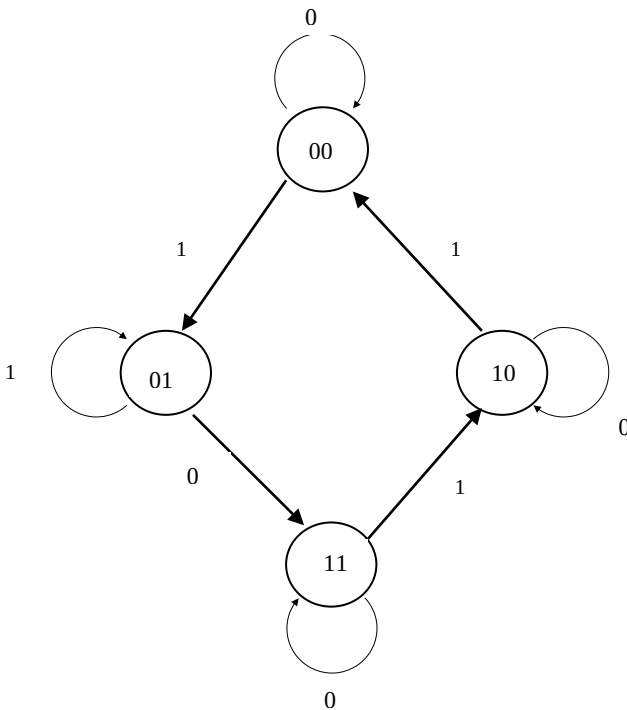
Implement the following Boolean functions using $3 \times 4 \times 2$ PLA.

$$F1(A, B, C) = \sum(3, 5, 6, 7)$$

$$F2(A, B, C) = \sum(0, 2, 4, 7)$$

- b) For the Boolean function $F(A, B, C) = A + B'C$: 7
 - i. Obtain the truth table
 - ii. Implement it using a MUX
 - iii. Implement it using a ROM

5. a) Realize the following state diagram into a sequential circuit using SR flipflop: 8



- b) Define flip flop. Explain the working of JK flip flop with its truth table and logic diagram. Also explain how it eliminates the problem encountered in SR flip flop. 7
6. a) Design 2-bit synchronous up counter by using JK Flipflop. 8
- b) Design a 4-bit arithmetic circuit which performs eight different arithmetic operations. 7
7. Write short notes on: (**Any two**) 2×5
- Priority Encoder
 - Status Register
 - Master-Slave flip-flop